

Patient Information

- **Name:** Jane Doe
- **Patient ID:** #9928374
- **DOB:** 10/12/1990
- **Gender:** Female

1. Complete Blood Count (CBC)

Parameter	Result	Reference Range	Units	Status
White Blood Cell (WBC)	7.4	4.5 – 11.0	$\times 10^3/\mu\text{L}$	Normal
Red Blood Cell (RBC)	3.8	4.2 – 5.4	$\times 10^6/\mu\text{L}$	Low
Hemoglobin (Hgb)	10.5	12.0 – 16.0	g/dL	Low
Hematocrit (Hct)	32.1	37.0 – 47.0	%	Low
Platelets	250	150 – 450	$\times 10^3/\mu\text{L}$	Normal

2. Blood Test Results (Metabolic Panel)

- **Glucose (Fasting):** 102 mg/dL (*High; Range: 70–99*)
- **Total Cholesterol:** 215 mg/dL (*High; Range: < 200*)
- **Vitamin D:** 22 ng/mL (*Low; Range: 30–100*)

3. Abnormal Values

- **Anemia Profile:** Red blood cell count, Hemoglobin, and Hematocrit are all below the standard clinical range.
- **Blood Sugar:** Fasting glucose shows mild elevation above the normal threshold.
- **Lipids:** Total cholesterol is slightly elevated.

4. Medical Conditions

- **Iron Deficiency Anemia:** Indicated by low Hgb and Hct levels.

- **Mild Vitamin D Deficiency:** Common in patients with limited sun exposure or dietary gaps.
- **Pre-diabetes (Borderline):** Fasting glucose suggests a need for monitoring.

5. Recommendations

- **Supplementation:** Consult a physician regarding Iron and Vitamin D3 supplements.
- **Dietary Adjustments:** Increase intake of leafy greens, red meat (or fortified plant-based alternatives), and reduce refined sugar intake.
- **Follow-up:** Repeat the CBC and Lipid panel in 3 months to monitor progress.
- **Exercise:** Incorporate 30 minutes of moderate activity 5 times per week.